Fabrication and Integration Process

1. Material Choice Justification

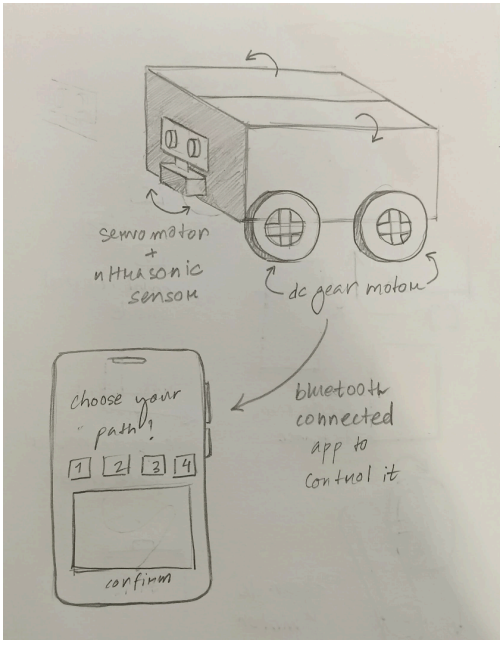
The base of the vehicle is made using a laser-cut MDF box that encloses the entire circuit and wiring. MDF was chosen mainly because it is strong and stable, so it can support the structure without bending, but at the same time it is not too heavy, which helps the motors run smoothly without extra load. Keeping the electronics inside this enclosure also protects them from damage and dust. Since MDF is opaque, it hides all the wiring, which makes the overall design look much cleaner and more organised instead of messy.

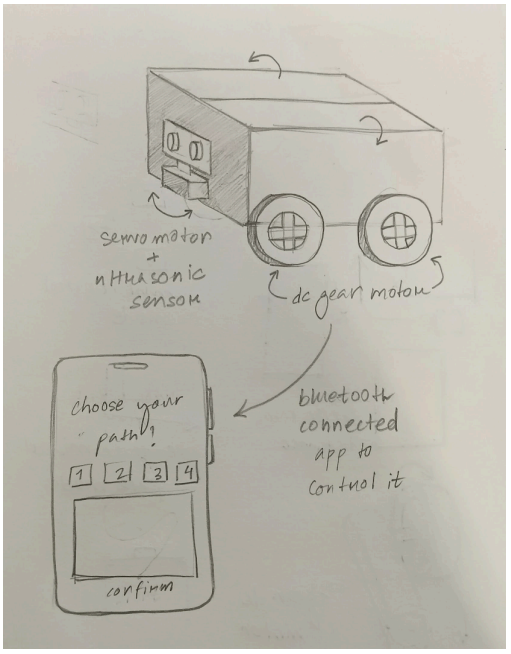
On top of this, a laser-cut acrylic box is used as the delivery compartment. This is the part where objects can be placed and transported. Acrylic was selected because it has a smooth and glossy finish, which gives the vehicle a more modern and visually appealing look. It also creates a nice contrast with the MDF base. Apart from appearance, acrylic is quite rigid and durable, so it can hold the weight of items without bending or cracking easily.

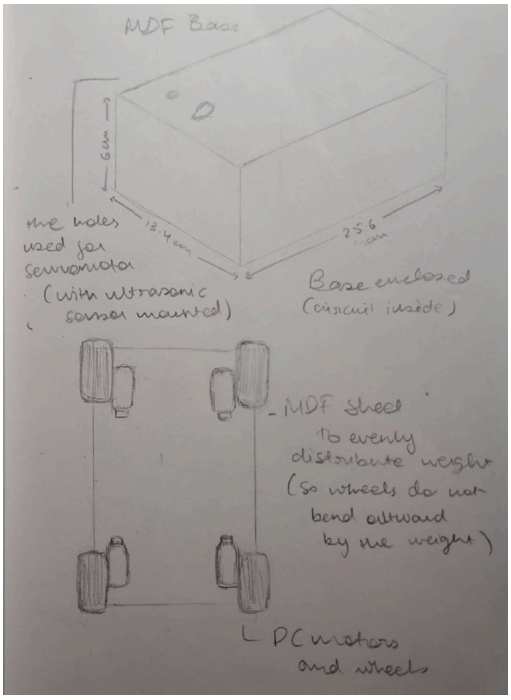
An additional MDF layer is placed beneath the DC motors. This acts as a support base that helps distribute the weight of both the MDF and acrylic boxes evenly across all four wheels. Without this support, the motors could experience uneven pressure, which might cause misalignment or unstable movement. By adding this layer, the vehicle remains balanced and moves more smoothly.

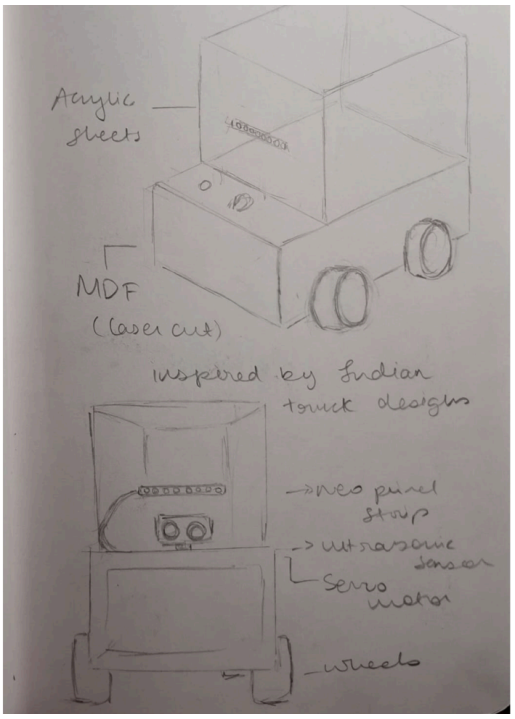
2. Fabrication Steps

1. First, we created digital design files for both the MDF base and the acrylic box, making sure the dimensions were accurate enough to fit all the components like the circuit, wiring, and delivery space properly.
2. These designs were then reviewed and approved. After making the required changes based on feedback, I arranged the MDF and acrylic sheets in the correct sizes and thicknesses needed for fabrication.
3. All the individual parts were then laser cut from their respective materials according to the final design files.
4. Before final assembly, we temporarily put together the boxes and chassis to test whether the motors were still able to move the vehicle properly with the added weight of the structure.
5. To improve stability, we added and fixed an extra MDF sheet below the motors. This helped distribute the load more evenly and prevented the wheels from tilting outward due to excess weight.
6. After completing all testing and making necessary adjustments, we permanently secured the circuit inside the MDF enclosure, fixed all the layers together, and assembled the full vehicle into its final working form.


servo motor
+
ultrasonic
sensor
dc gear motor
choose your path!
1
2
3
4
confirm
bluetooth
connected
app to
control it


servo motor
+
ultrasonic
sensor
dc gear motor
choose your path!
1
2
3
4
confirm
bluetooth
connected
app to
control it


MDF Base
6cm
12.4cm
25.6 cm
the holes used for servomotor (with ultrasonic sensor mounted)
Base enclosed (circuit inside)
MDF sheet to evenly distribute weight (so wheels do not bend outward by the weight)
DC motors and wheels


Acylic sheets
MDF
(laser cut)
inspired by Indian truck designs
neo pixel strip
ultrasonic sensor
servo motor
wheels